2025-26 S1 – BIO220 PROGRAM

Week	Lecture	Tutorials and Labs	Coursework
Week 1	Introduction to computational Biology.		
Week 2	Charges and Coulomb law. Electrostatic potential.		
Week 3	Atoms and molecules. The Schrodinger equation		
Week 4	Molecular mechanics. Force Fields.		Coursework 1 (in class test): Molecular Physics.
Week 5	Structure of proteins and nucleic acids.	Tutorial and Lab: Molecular visualization with VMD.	
Week 6	Structural models and protein folding. Homology modeling and Alphafold.	Tutorial and Lab: Homology modeling.	Coursework 2 (laboratory): Homology modeling and molecular visualization.
Week 7	Molecular docking.		
Week 8	Energy minimization. Normal modes analysis.		Coursework 2 deadline
Week 9	Molecular surfaces and electrostatics.	Tutorial and Lab: Molecular docking, normal modes.	
Week 10	Free energy and solvation.		
Week 11	Small molecules and chemical spaces. Chemoinformatics.	Tutorial and lab: Protein/Ligand modeling.	Coursework 3 (laboratory): Protein/Ligand modeling.
Week 12	Drug libraries, screening and selection.		
Week 13	Recap.		Coursework 3 deadline

MOLECULAR DOCKING

Measurement units for energy

- KJ/mol (kilojoules per mole)
  - SI unit for energy per mole
  - 1 kJ/mol = 0.239 kcal/mol

From the definition of work: $1\ J = 1\ Nm$
Gas constant $R = 8.314\ J/(mol\ K)$

- $Kcal/mol$ (kilocalories per mole)
  - Non-SI unit for energy per mole
  - 1 kcal/mol = 4.184 kJ/mol

$1\ kcal$: amount of heat required to increase 1 K the temperature of 1 Kg of water

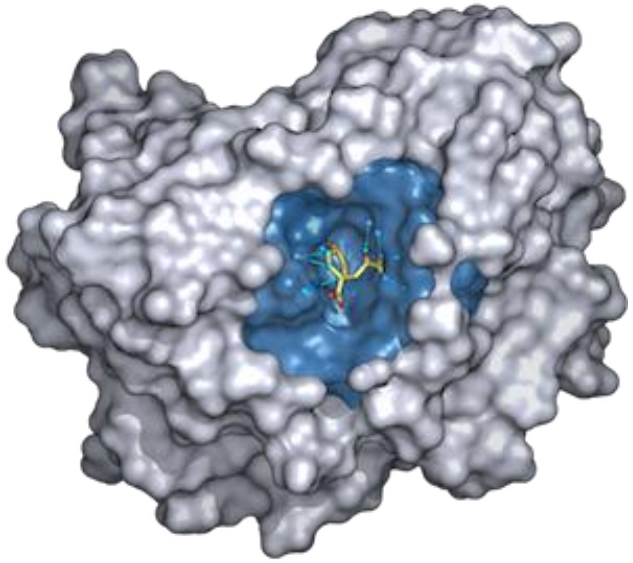
- K_BT (Boltzmann constant × Temperature)
  - Unit for thermal energy
  - At room temperature (300K), 1 kT ≈ 0.592 kcal/mol

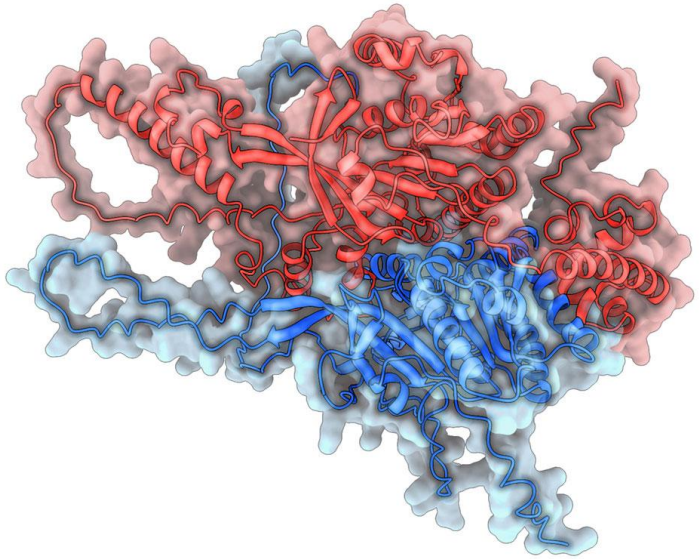
Gives a measure of the average kinetic energy at a certain temperature.
Boltzmann constant: $k_B = \frac{R}{N_A} = 1.38 \cdot 10^{-23} J/K$

- eV (electron-volt)
  - Unit for energy
  - Potential energy of an electron feeling a potential of 1V

- Conversion factors:
  - 1 kJ/mol = 0.239 kcal/mol
  - 1 $kcal/mol$ = 4.184 kJ/mol
  - 1 K_BT (at 300K) ≈ 0.592 kcal/mol ≈ 2.48 kJ/mol
  - 1 K_BT (at 300K) ≈ 26 meV ($26\ \cdot 10^{-3}\ eV$)

Binding between molecules





Biological molecules **interact and bind** to perform essential functions (e.g., enzyme activity, signaling, structural stability).

Types of Molecular Binding:
Protein-ligand binding (e.g., enzyme-substrate, drug-target interactions)
Protein-protein interactions (e.g., signaling pathways, antibody-antigen)
Protein-DNA/RNA binding (e.g., transcription factors, ribosomes)
Membrane-receptor binding (e.g., hormone-receptor, neurotransmitters)

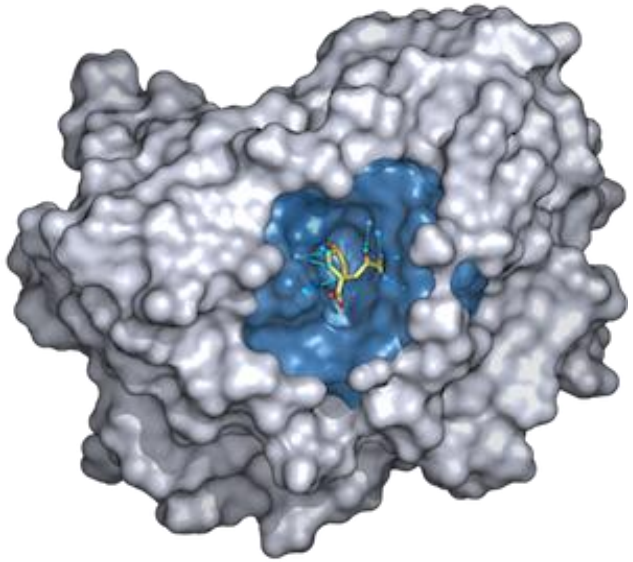
Binding is mediated by non-covalent interactions:
Hydrogen bonds (specificity and recognition, e.g., DNA base pairing).
Electrostatic interactions (salt bridges, charged side chains).
Van der Waals forces (stabilization in molecular docking).
Hydrophobic interactions (key for protein folding and drug binding).
Binding Affinity & Specificity:

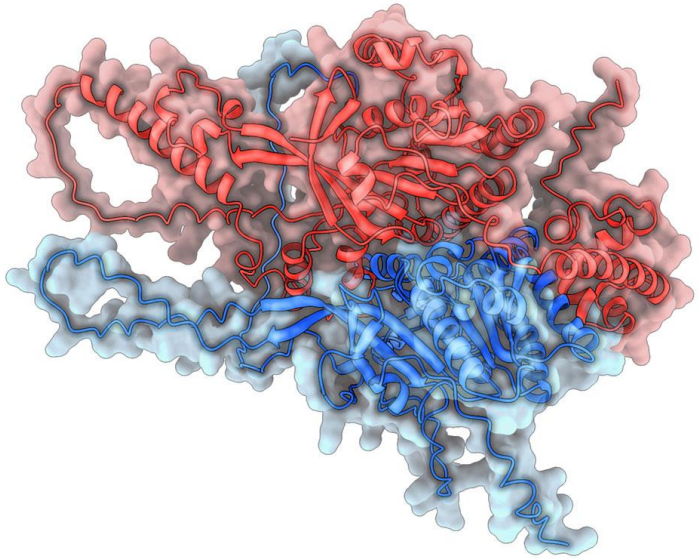
Strength of interaction depends on binding affinity (measured by K_D).
High specificity ensures correct molecular recognition (e.g., enzymes recognizing substrates).

Dynamic Nature of Binding:
Binding is often reversible, allowing biological regulation.
Some interactions lead to allosteric effects (binding at one site affects another).

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Molecular Docking





There are much less known structures of molecular interaction compared to isolated molecules.

Molecular docking is mainly used in two cases: modeling of Protein-Ligand interaction or Protein-Protein interaction.

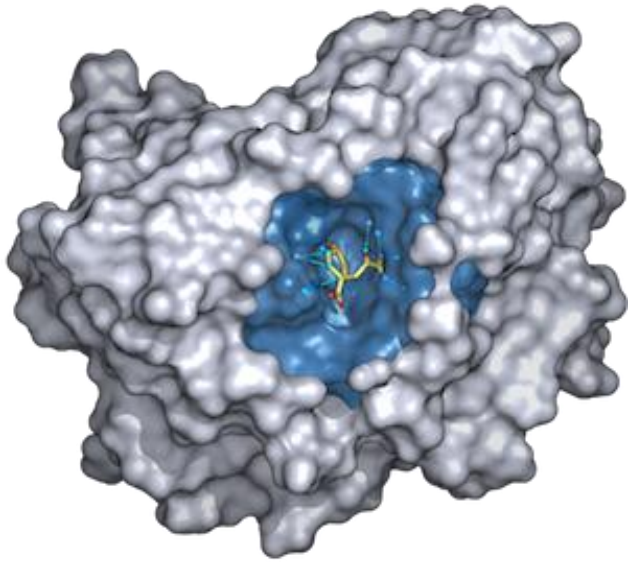
Protein-Ligand interaction is often used in drug design.

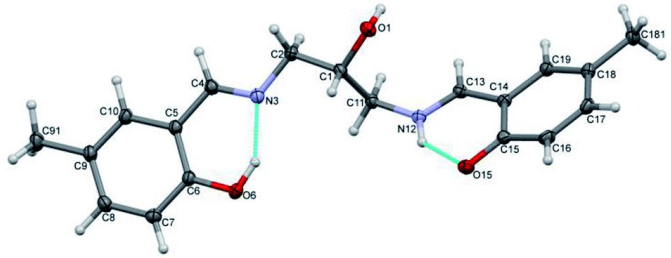
Protein-Protein interaction is important to understand the interaction between proteins in the cell or to design new biological drugs.

While the two problems are similar in nature, the differences between small molecules and proteins are so important that it makes sense to treat them as separate problems.

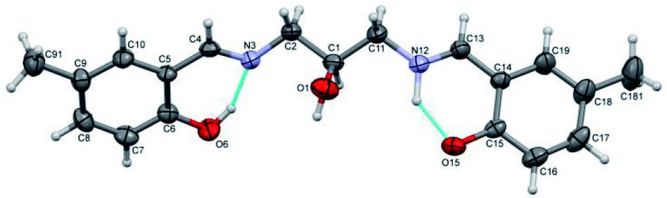
We have different tools and software to deal with the two different problems.

Molecular Docking





(a)

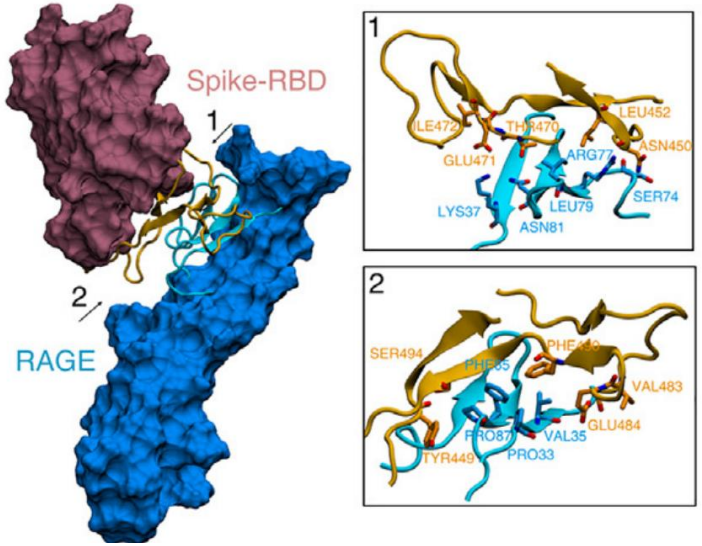


(b)

Protein-Ligand interactions

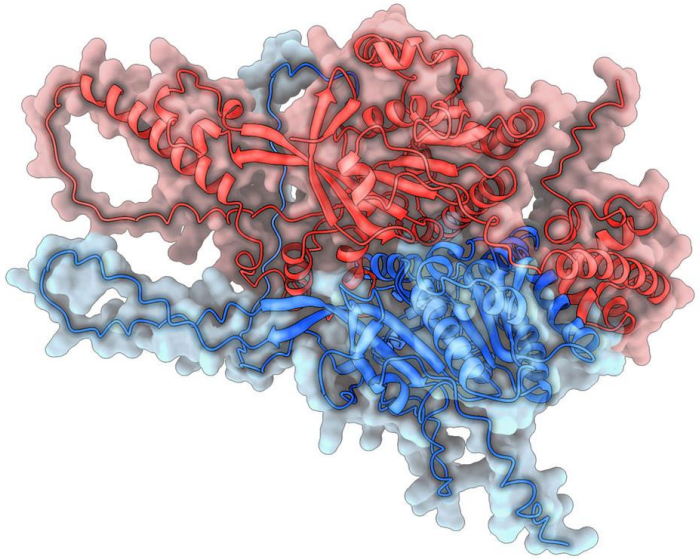
- Ligands are usually small and more rigid than proteins (few degrees of freedom).
- Small molecules usually bind to pockets (cavities).
- Interactions are often mediated by hydrophobic contributions.
- In many practical cases, a large number of ligands need to be tested (screened).

Molecular Docking



Protein-Ligand interactions

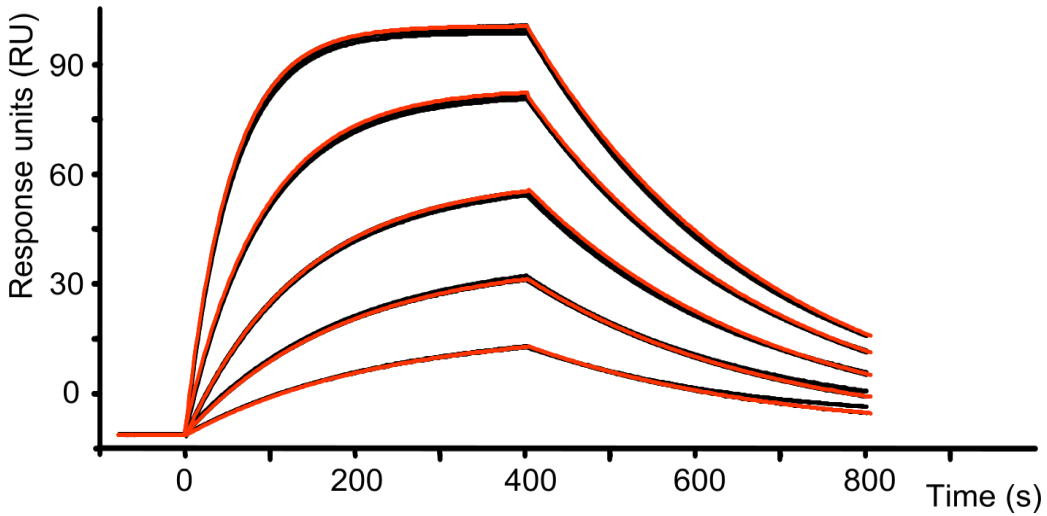
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- Interactions are often mediated by hydrophobic contributions.
- In many practical cases, a large number of ligands need to be tested (screened).



Protein-Protein interactions

- Proteins are large and can have important configuration changes upon binding.
- The interaction between the proteins happens on their surface.
- It is often mediated by hydrophilic residues and by surface complementarity.
- The interacting proteins are usually known (no need for screening).

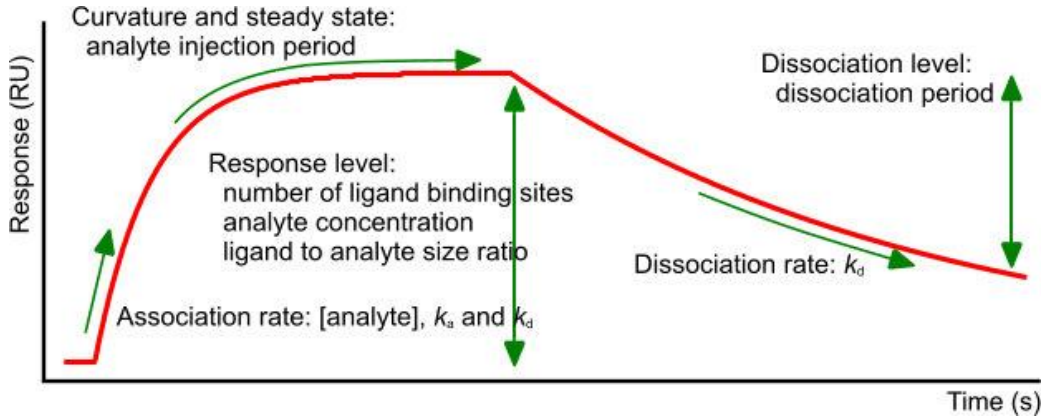
Molecular binding is a dynamical process

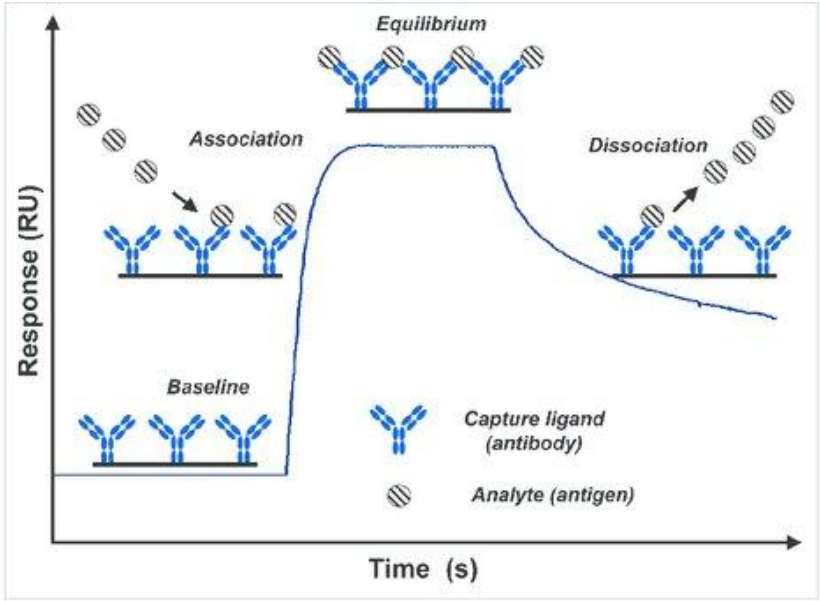


Typical SPR (Surface Plasom Resonance) experiment.

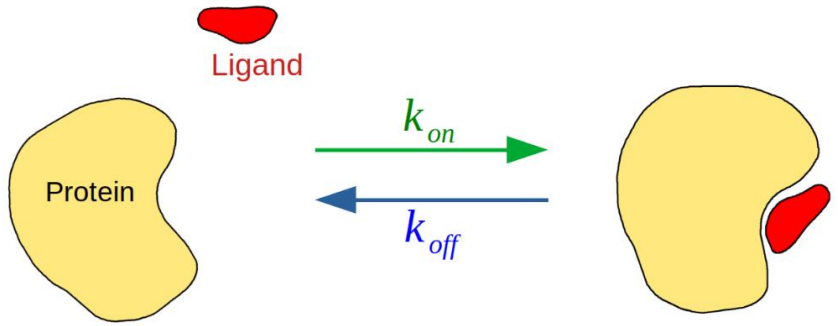
SPR measure binding and unbinding of two molecules as a function of the concentration.

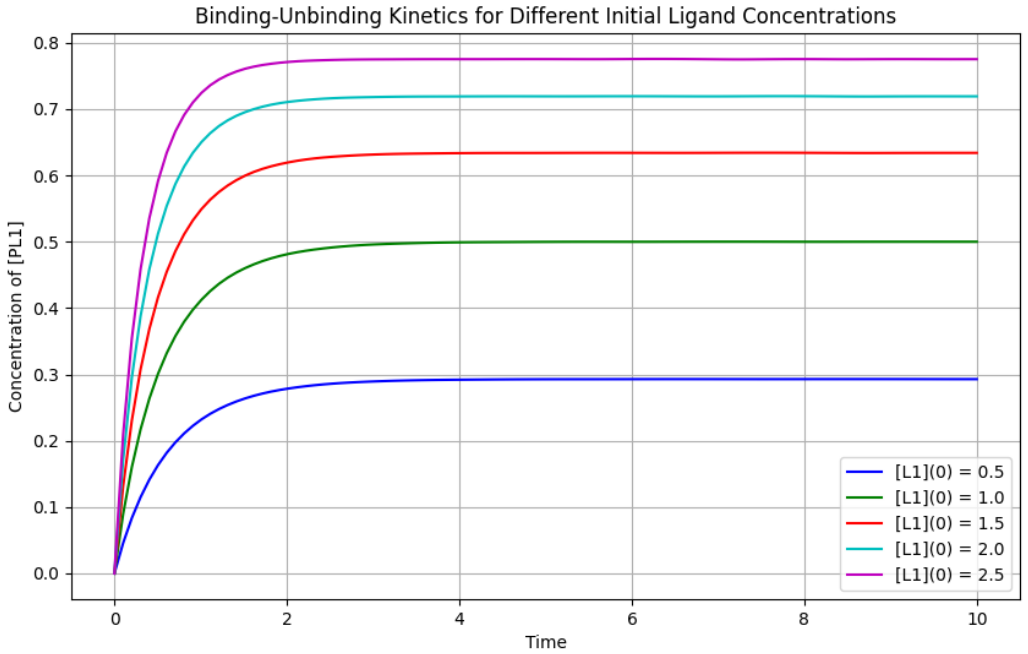
The ratio between the K_{on} and K_{off} is the equilibrium constant (K_D).





Molecular binding is a dynamical process





Typical SPR (Surface Plasom Resonance) experiment.

SPR measure binding and unbinding of two molecules as a function of the concentration.

The ratio between the K_{on} and K_{off} is the equilibrium constant (K_D).

Microscopically, binding works like a chemical reaction, with a dynamical equilibrium:

$$P + L_1 \leftrightharpoons PL_1$$

$$\frac{d[P]}{dt} = -k_{on}[P][L_1] + k_{off}[PL_1]$$

$$\frac{d[L_1]}{dt} = -k_{on}[P][L_1] + k_{off}[PL_1]$$

$$\frac{d[PL_1]}{dt} = k_{on}[P][L_1] - k_{off}[PL_1]$$

Competitive binding

$$P + L_1 \leftrightharpoons PL_1$$

$$P + L_2 \leftrightharpoons PL_2$$

$$\frac{d[P]}{dt} = -k_{on1}[P][L_1] + k_{off1}[PL_1] - k_{on2}[P][L_1] + k_{off2}[PL_2]$$

$$\frac{d[L_1]}{dt} = -k_{on1}[P][L_1] + k_{off1}[PL_1]$$

$$\frac{d[L_2]}{dt} = -k_{on2}[P][L_2] + k_{off2}[PL_2]$$

$$\frac{d[PL_1]}{dt} = k_{on1}[P][L_1] - k_{off1}[PL_1]$$

$$\frac{d[PL_2]}{dt} = k_{on1}[P][L_2] - k_{off1}[PL_2]$$

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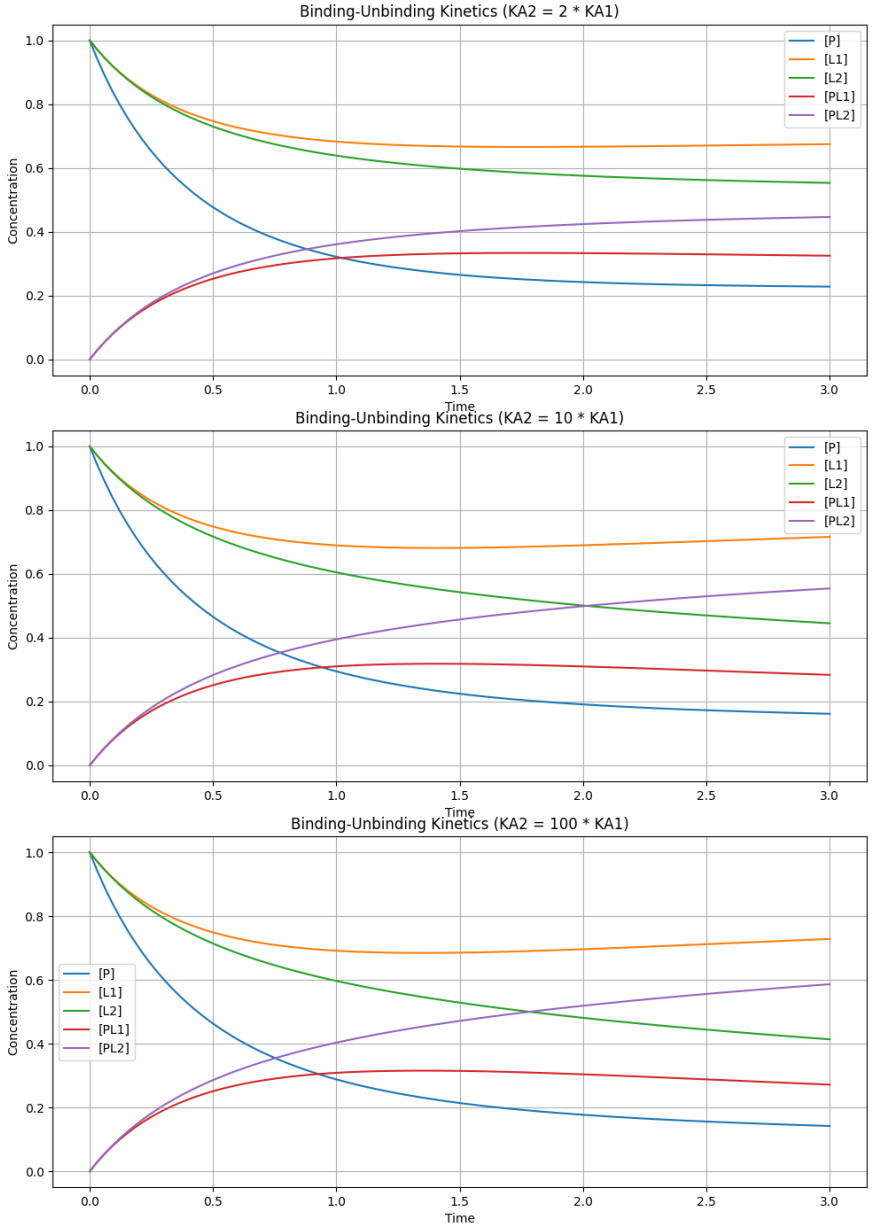
$$\frac{d[P]}{dt} = -k_{on1}[P][L_1] + k_{off1}[PL_1] - k_{on2}[P][L_1] + k_{off2}[PL_2]$$

$$\frac{d[L_1]}{dt} = -k_{on1}[P][L_1] + k_{off1}[PL_1]$$

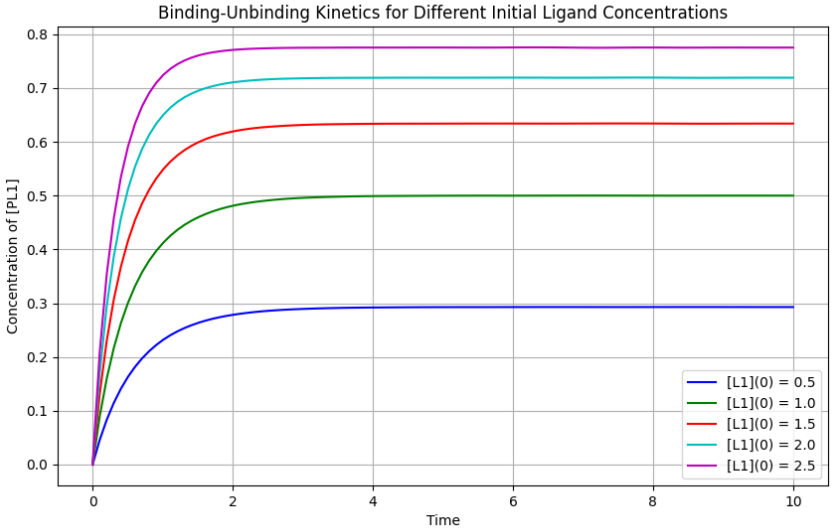
$$\frac{d[L_2]}{dt} = -k_{on2}[P][L_2] + k_{off2}[PL_2]$$

$$\frac{d[PL_1]}{dt} = k_{on1}[P][L_1] - k_{off1}[PL_1]$$

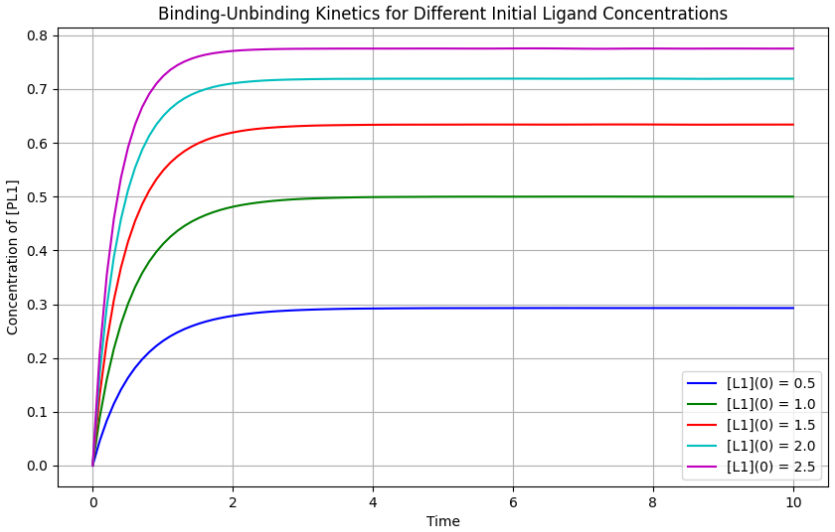
$$\frac{d[PL_2]}{dt} = k_{on1}[P][L_2] - k_{off1}[PL_2]$$

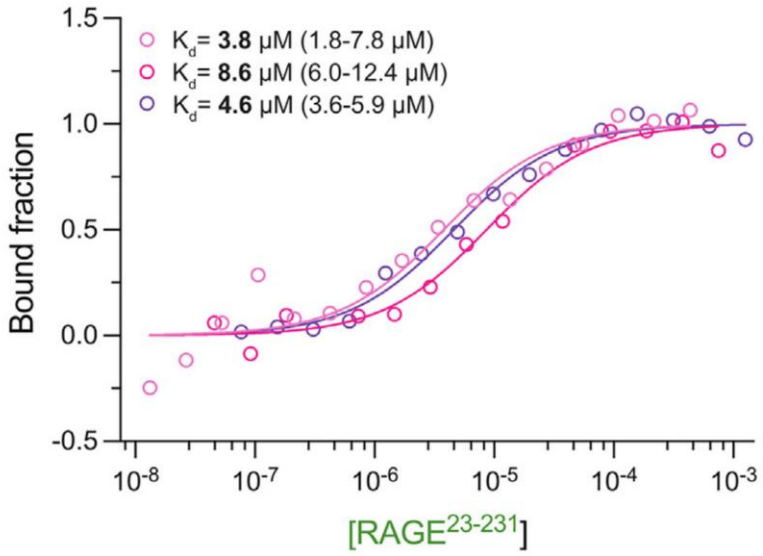


Binding affinity



Binding affinity





Binding experiments measure the fraction of target receptors occupied at equilibrium across a range of ligand concentrations.

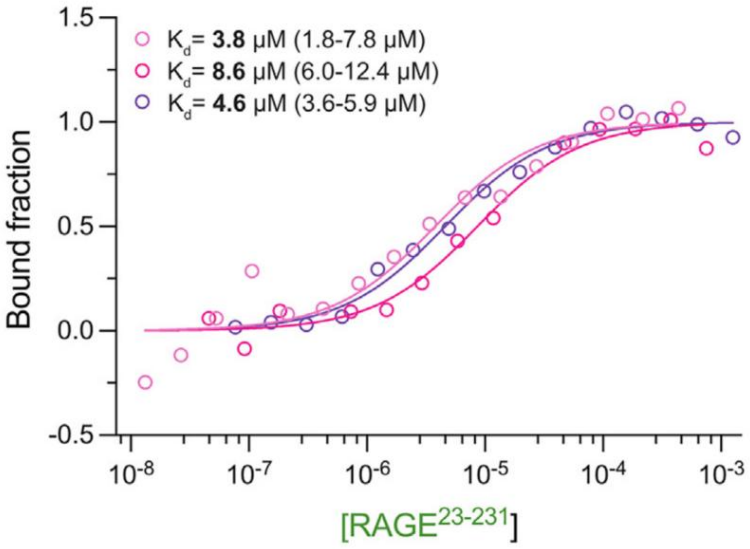
Plotting equilibrium occupancy against ligand concentration generates a characteristic sigmoid saturation curve.

The inflection point of this sigmoid (where 50% of receptors are bound) directly corresponds to the ligand concentration equal to the K_D (see next slide).

Multiple measurements across different concentrations allow fitting to the binding equation, yielding K_D values with statistical confidence intervals.

This equilibrium saturation approach provides a direct experimental readout of binding affinity by identifying the concentration needed for half-maximal target occupancy.

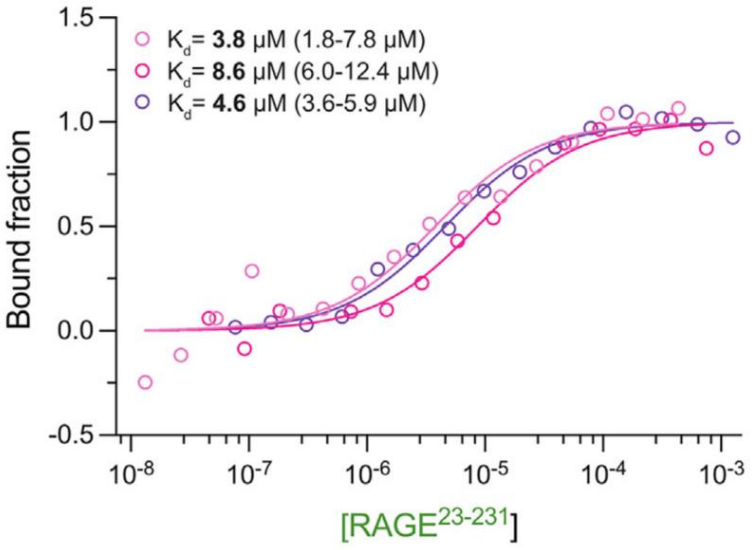
Binding Affinity and Binding Free Energy



Let us assume we have a generic chemical reaction: $A_xB_y \rightleftharpoons xA + yB$; we call *dissociation constant* the quantity:

$$K_D = \frac{[A]^x[B]^y}{[A_xB_y]}$$

Binding Affinity and Binding Free Energy



The binding affinity K_A is the reciprocal of the dissociation constant:

$$K_A = \frac{1}{K_D}$$

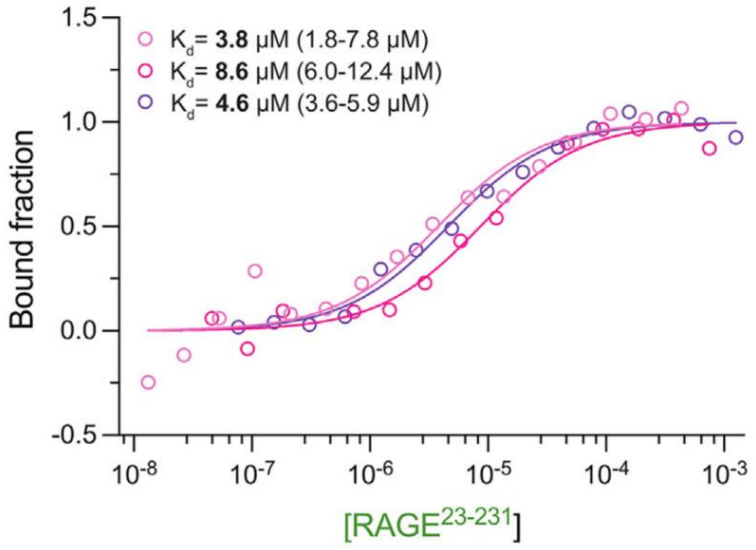
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If the reaction is 1:1 ($x = 1$, $y = 1$), the dissociation constant represents the concentration at which half of the molecules are interacting:

When $[B] = [AB]$, $K_D = [A]$: $\frac{[AB]}{[B] + [AB]} = \frac{1}{2}$

Binding Affinity and Binding Free Energy



The binding affinity K_A is the reciprocal of the dissociation constant:

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The dissociation constant is linked to the binding free energy ΔG (difference of free energy between the bound and unbound state) by the following equation:

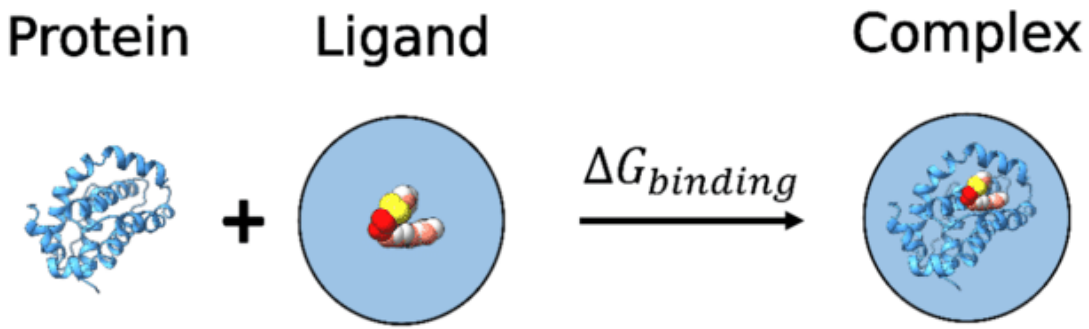
$$\Delta G = RTlnK_D \qquad K_D = e^{\Delta G/RT}$$

Energy measured in $KJ/mol \cdot K$

$$K_D = e^{\Delta G/K_BT}$$

Energy measured in K_BT

Binding Affinity and Binding Free Energy

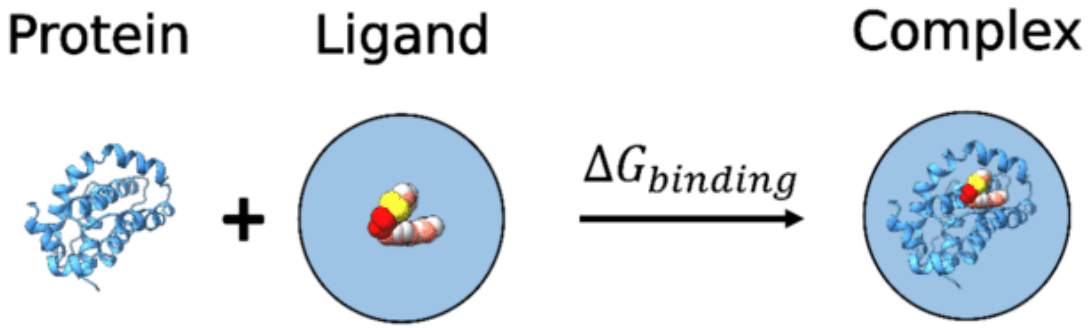


ΔG is the difference of free energy between the bound and unbound state:

$$\Delta G = \Delta H - T\Delta S = G_B - G_U = U_B - U_U - TS_B - TS_U$$

$\Delta H = \Delta U + p\Delta V = \Delta U$ if no change in volume happens

Binding Affinity and Binding Free Energy



ΔG is the difference of free energy between the bound and unbound state:

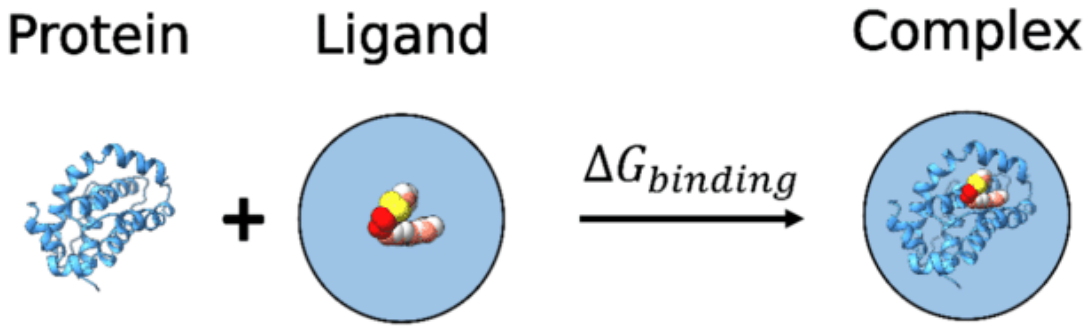
$$\Delta G = \Delta H - T\Delta S = G_B - G_U = U_B - U_U - TS_B - TS_U$$

$$\Delta H = \Delta U + p\Delta V = \Delta U$$ if no change in volume happens

ΔH : Energy changes associated with breaking and forming intermolecular interactions, (hydrogen bonds, van der Waals interactions, and electrostatic interactions), during the binding. Can be favorable or unfavorable.

Energy changes (usually unfavorable) due to the loss of solvation of the proteins and ligands.

Binding Affinity and Binding Free Energy



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Energy changes (usually unfavorable) due to the loss of solvation of the proteins and ligands.

ΔS : change in the configurational space accessible to the proteins and ligands upon binding (usually unfavorable).

Release or reorganization of water molecules from the binding interfaces (can contribute favorably)

Upon binding both enthalpy and entropy can decrease, but there are no general rule.

Binding Affinity and Binding Free Energy

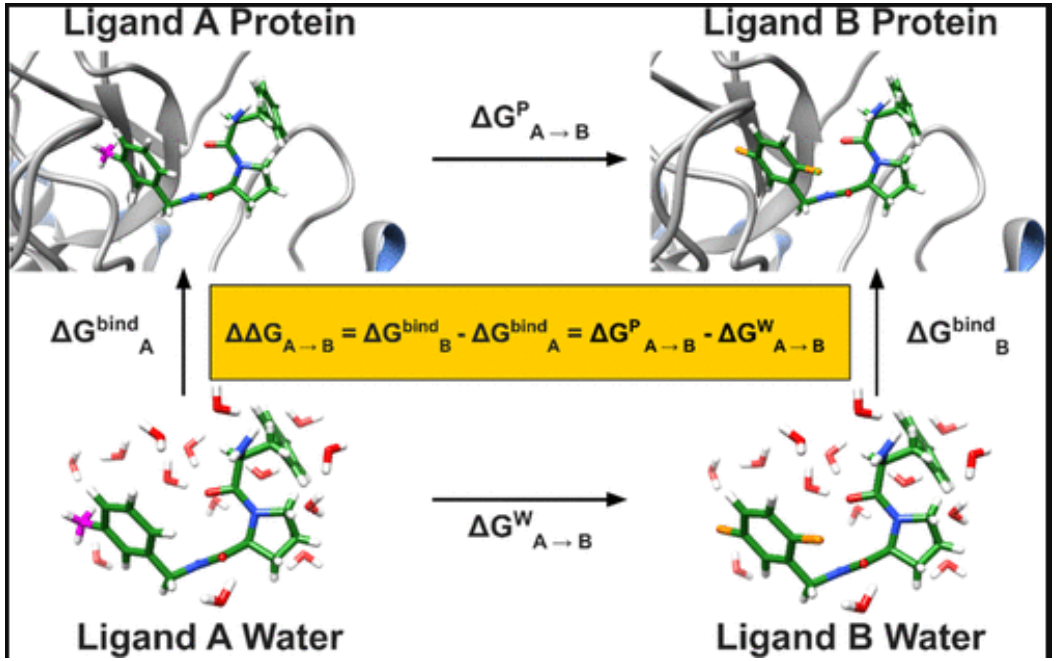
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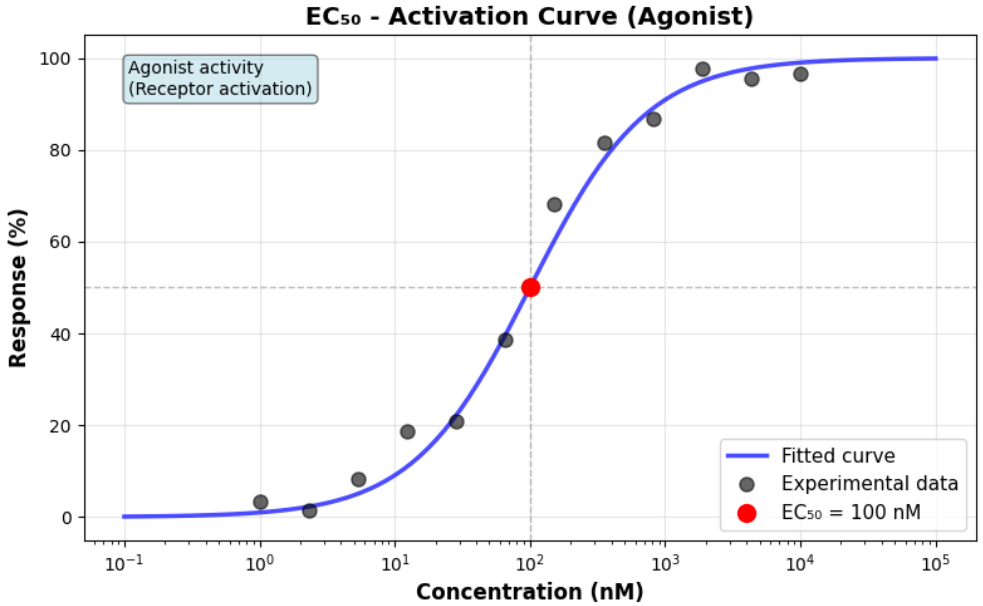
If we want to compare multiple ligands, we can calculate $\Delta\Delta G = \Delta G_B - \Delta G_A$, i.e. the difference between the two binding affinities

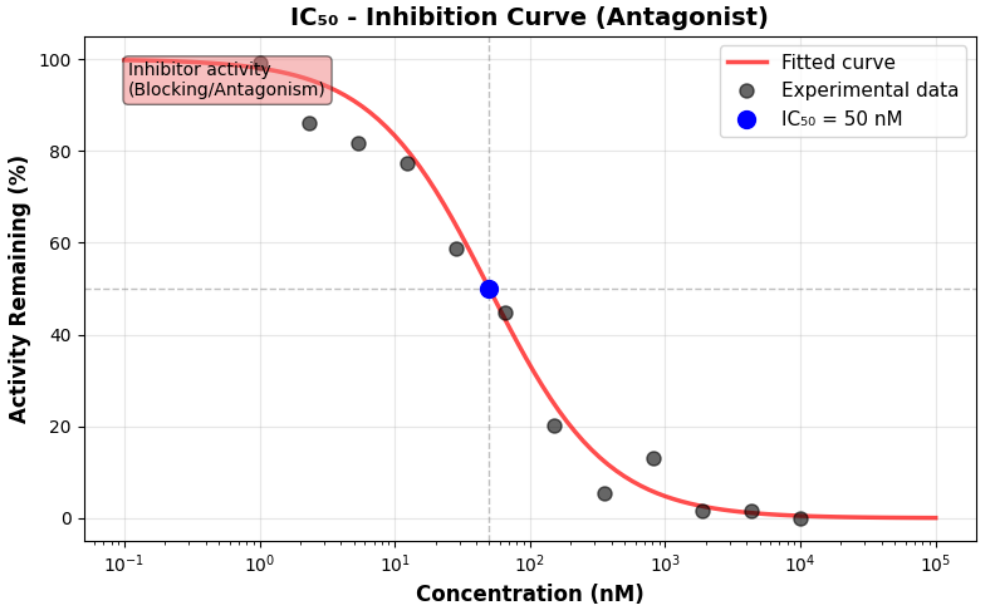
$$\frac{K_{DA}}{K_{DB}} = e^{(\Delta G_A - \Delta G_B)/K_B T} = e^{-\Delta\Delta G/K_B T}$$

If $\Delta\Delta G < 0$ B is a better binder than A
(lower binding energy, better binding affinity)



Binding Affinity and Binding Free Energy





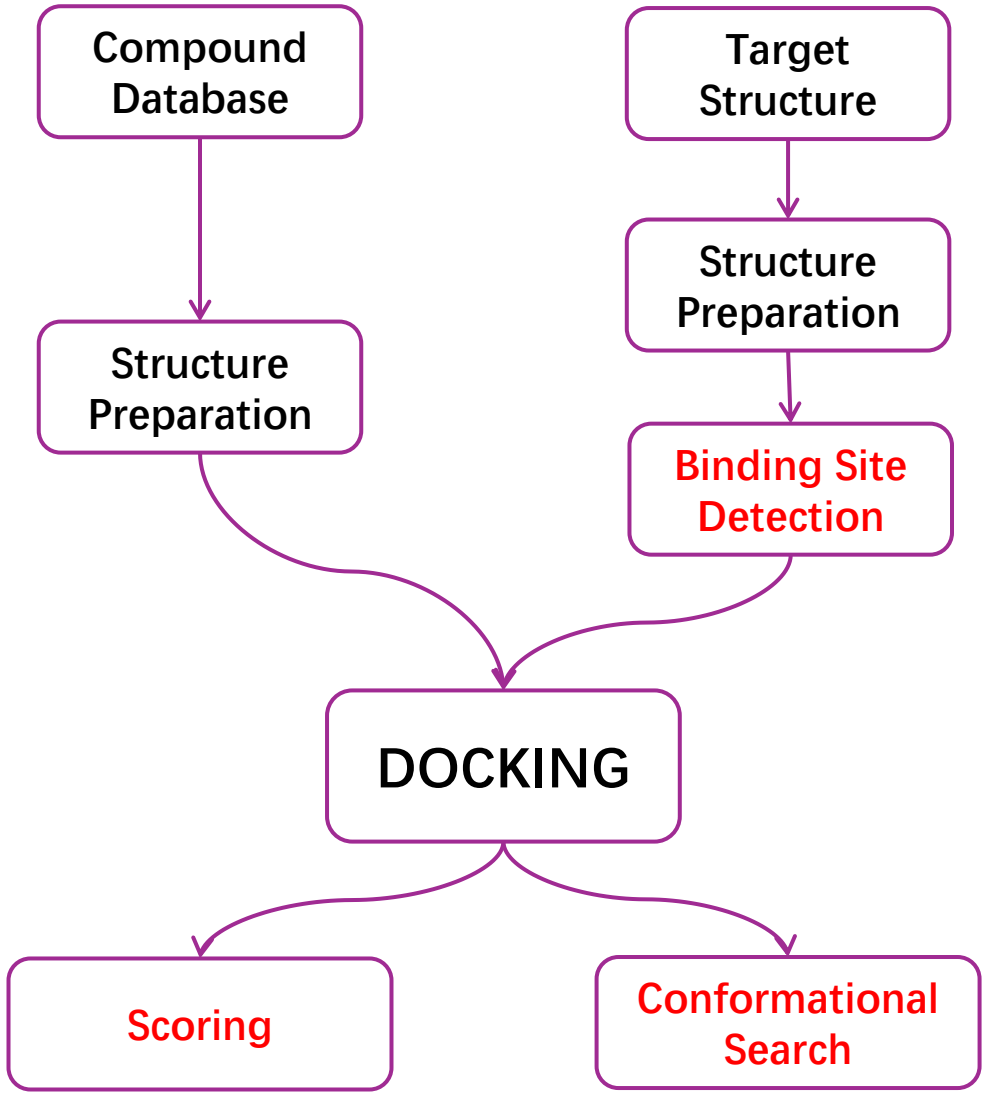
In biological and pharmaceutical applications, KD is often substituted by two other quantities.

EC_{50} - Half-maximal Effective Concentration. This measures functional potency. It is the concentration of a drug needed to produce 50% of its maximum agonistic (stimulatory) effect.

IC_{50} - Half-maximal Inhibitory Concentration. It is the concentration of an antagonist or inhibitor needed to block a specific biological function or response by 50%.

IC_{50} and EC_{50} are more commonly used in drug discovery because they measure the actual functional effect of the drug (like inhibiting cell growth or activating a pathway) in a more complex biological system, such as a cell-based assay.

Docking procedures and strategies



Binding Site Detection

- Identify the possible binding site from the literature (list of known binding sites) or from structural information.

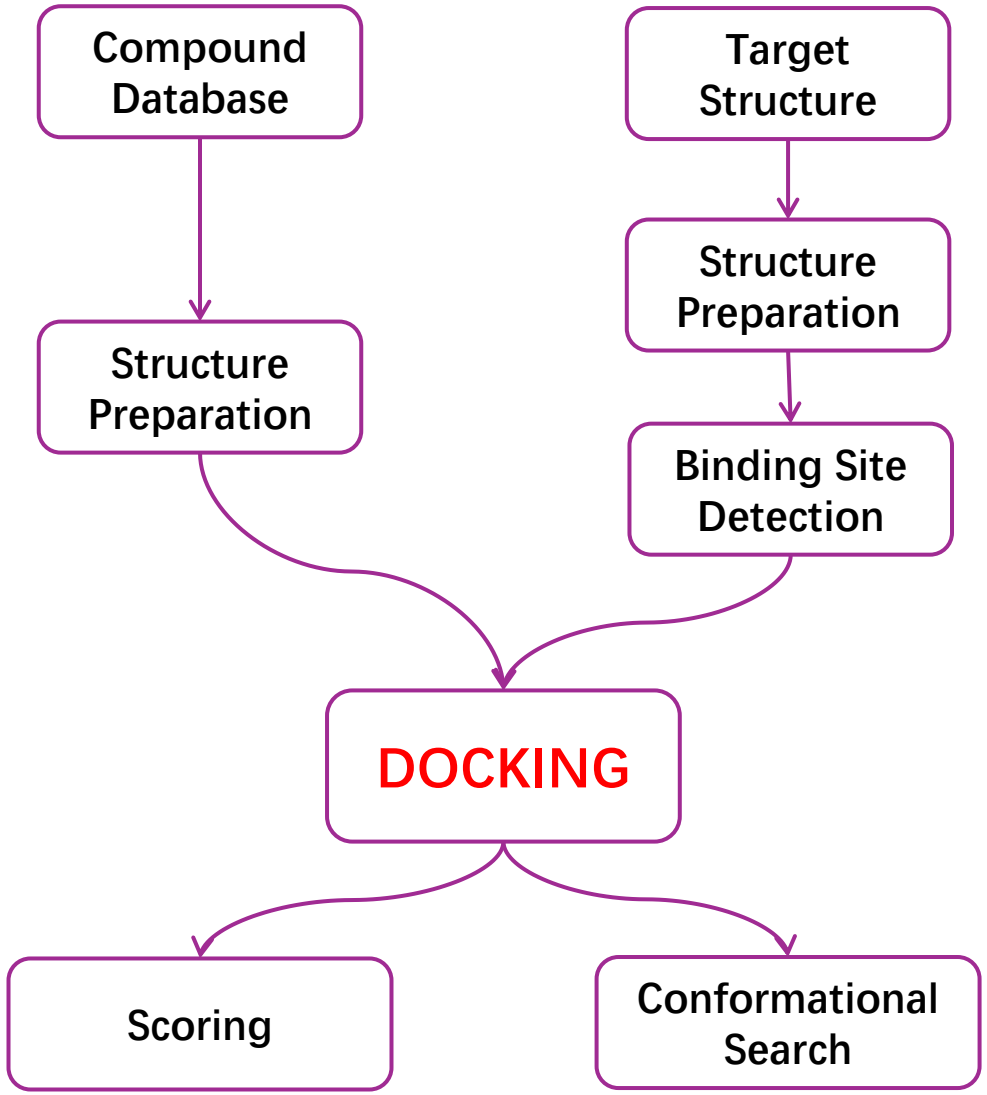
Scoring

- Evaluates the quality of the docking and allow to rank different results.
- It generally uses approximate calculation and can roughly been interpreted as binding energy.

Conformational Search

- Further analysis can be necessary to improve the quality of the results.
- It is possible to study the system dynamics to gain insight on the quality and stability of the docking.
- Usually requires simulations (see next lesson).

Docking procedures and strategies



Rigid docking

- Both molecules are assumed rigid in orientation
- Explore the best matching with minimized energy of the system
- Simple and less computational cost
- Suits for protein-protein, protein-nucleic acid

Semi-flexible docking

- Ligand – flexible, target - rigid
- Larger but reasonable computational cost
- Suits for protein-small molecule

Flexible docking

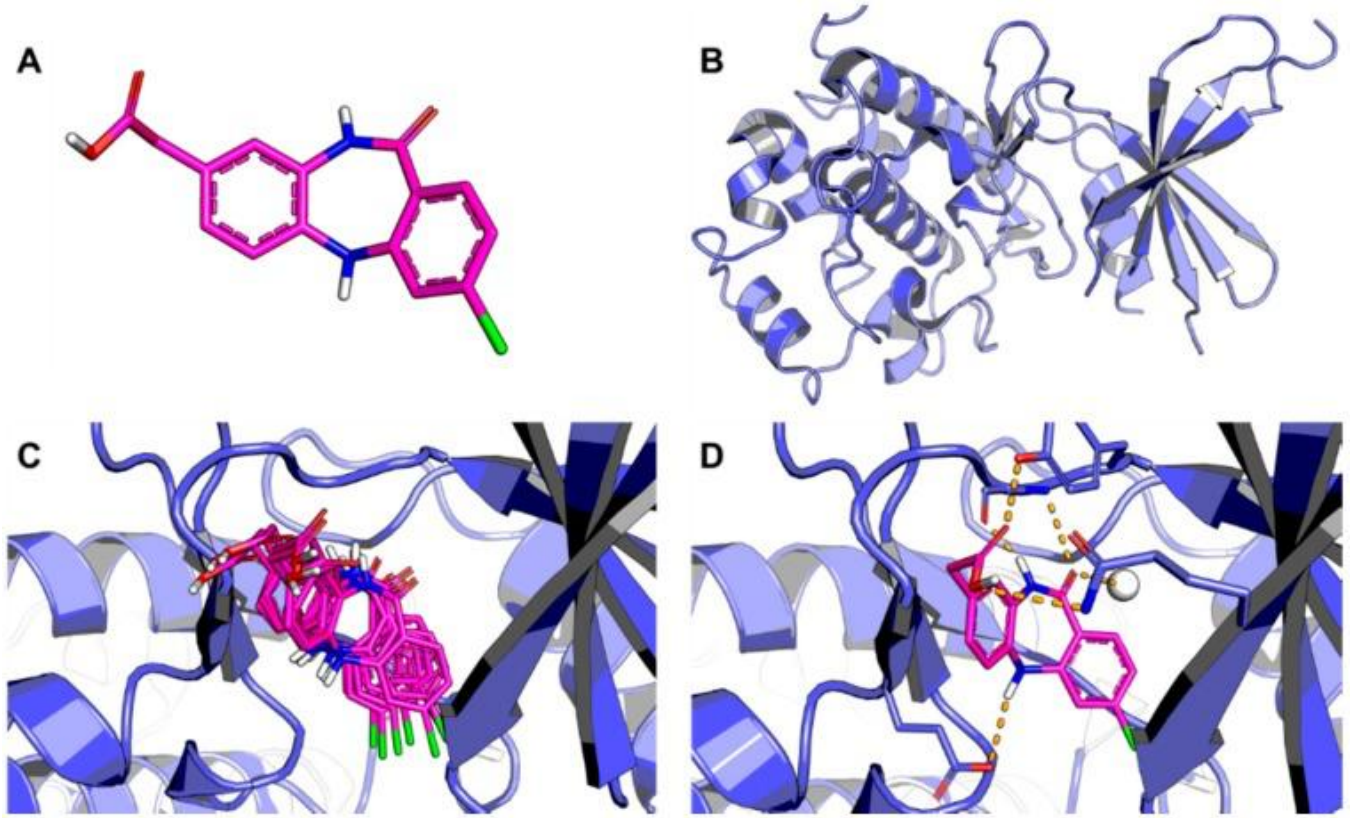
- Both molecules - flexible
- Tremendous computational cost
- Suits for accurate examination of sophisticated interaction

Rigid Docking

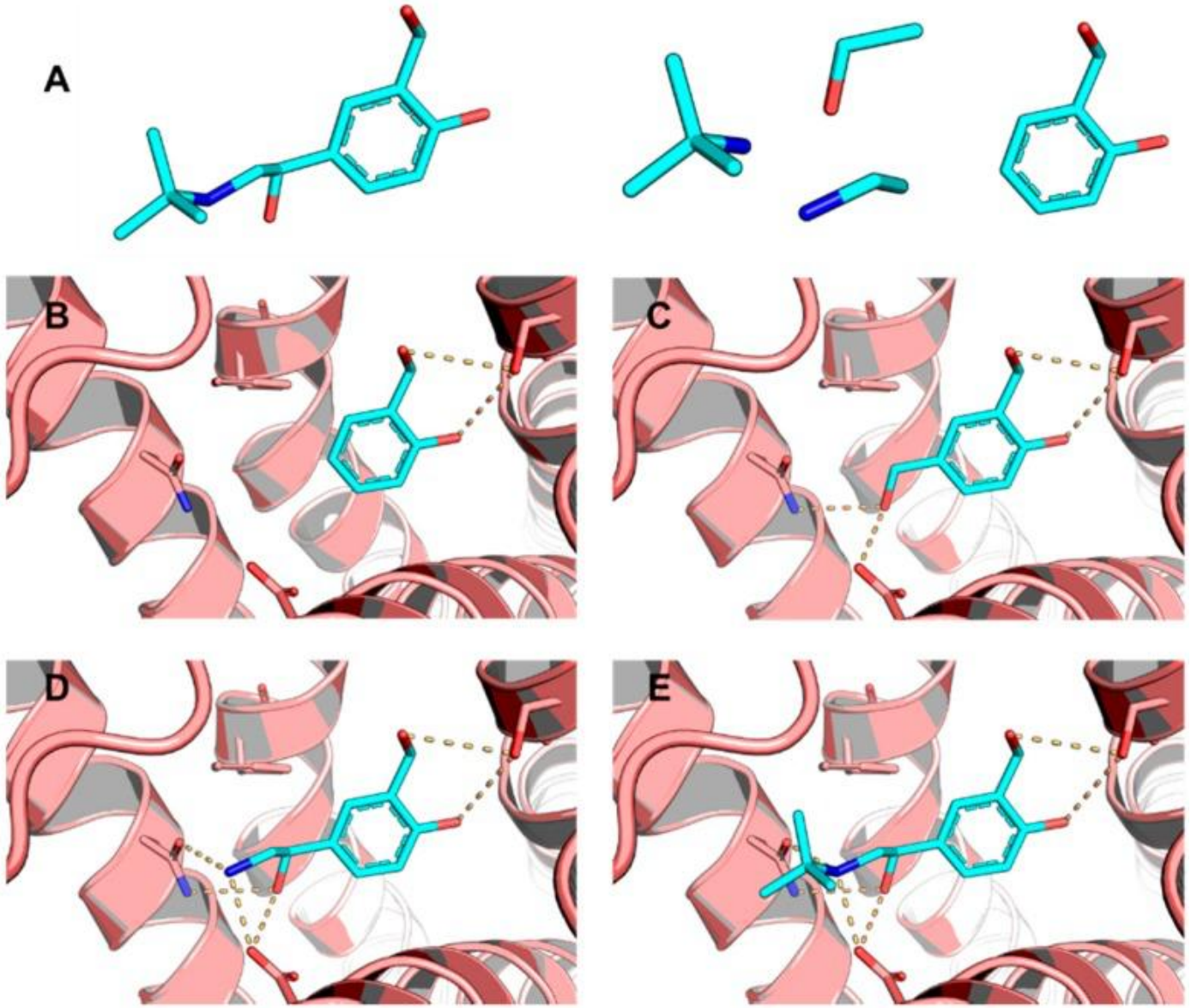
Outline of the molecular docking process. [**A**] 3D structure of the ligand; [**B**] 3D structure of the receptor; [**C**] The ligand is docked into the binding cavity of the receptor and the putative conformations are explored; [**D**] The most likely binding conformation and the corresponding intermolecular interactions are identified.

Notes: *The protein backbone is represented as a cartoon. The ligand (carbon in magenta) and active site residues (carbon in blue) are shown in stick representation. Water is shown as a white sphere and hydrogen bonds are indicated as dashed lines.*

Ferreira et al. Molecules. 2015 Jul; 20(7): 13384–13421.



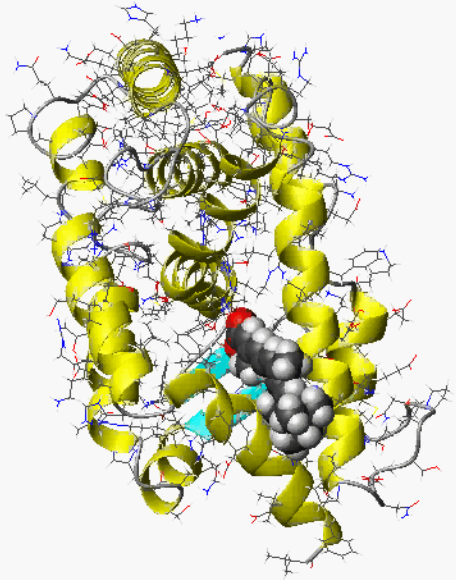
Semi Flexible Docking

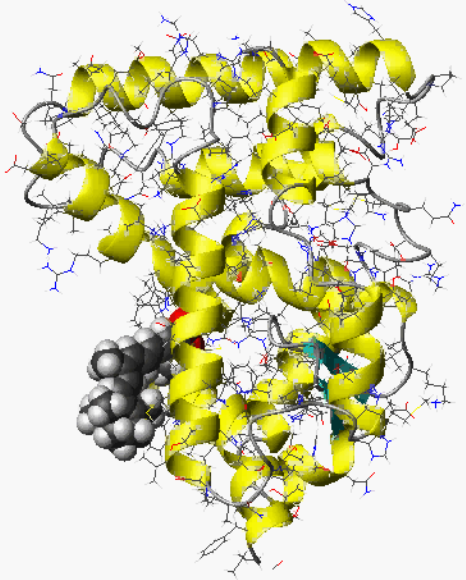


The incremental construction method. [**A**] The ligand (stick representation, carbon in cyan) is broken into several fragments; [**B**] The anchor fragment is docked in the binding site of the molecular target (cartoon representation, carbon in salmon); [**C**] The next fragment is docked after the anchor fragment; [**D** and **E**] The other fragments are docked sequentially to construct the entire ligand in its binding conformation. Residues in the active site are shown in stick representation (carbon in salmon). Hydrogen bonds are indicated as dashed lines.

Ferreira et al. Molecules. 2015 Jul; 20(7): 13384–13421.

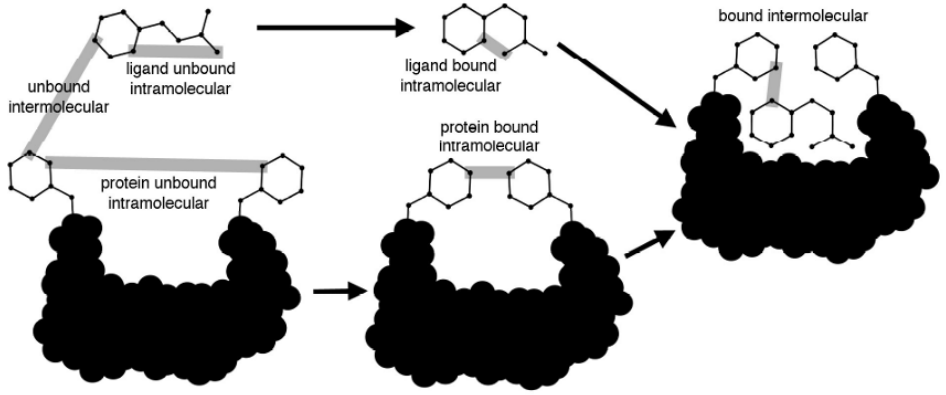
Semi Flexible Docking





Flexibility and solvent effects. Difficulties experienced in scoring and ranking compounds mainly arise from restraints and approximations applied to conformational flexibility and solvation effects. The dynamic nature of proteins and ligands in a physiological setting allows them to adopt a considerable number of different conformations. However, incorporating molecular flexibility in virtual screening of large chemical libraries is not feasible due to high computational costs. To solve this, different strategies that can partially account for protein flexibility in docking calculations are being used in various software packages: [**A**] 'Soft scoring' reduces steric energy penalties to permit some degree of overlap between ligand and target atoms. [**B**] Ensemble-based approach exploits multiple receptor structures that are experimentally determined (X-ray crystallography or NMR) or computationally produced using molecular dynamics. [**C**] Induced fit approach is based on the principle of induced fit effects that take place upon binding of a ligand to a receptor. [**D**] Molecular simulation based on force fields depicting intra- and intermolecular interactions between receptor and ligand is also possible.

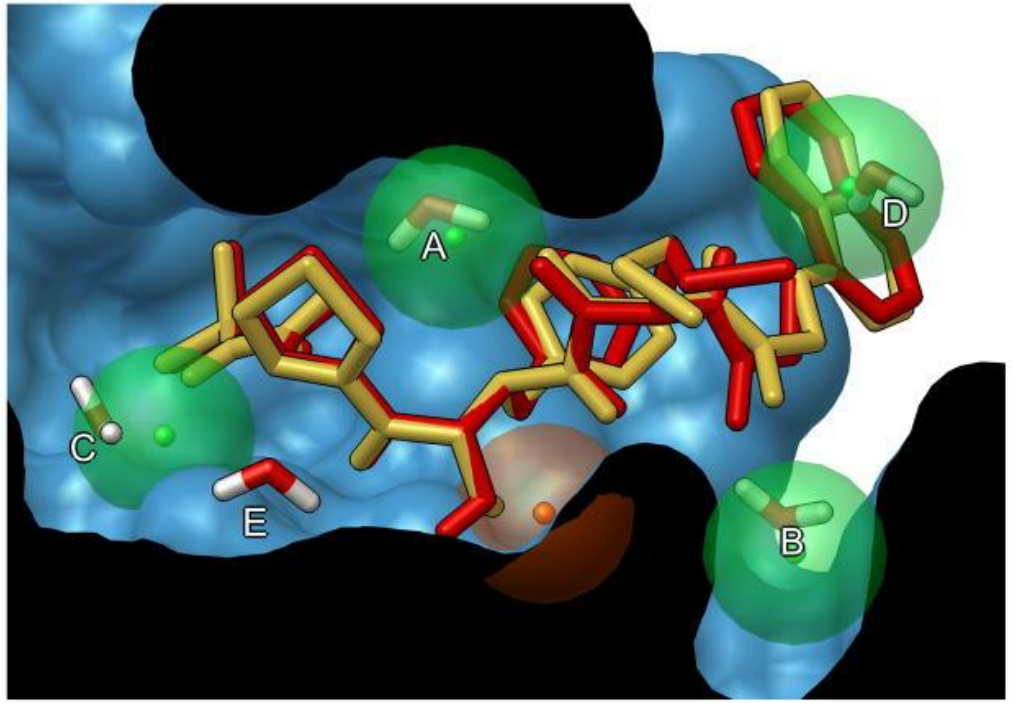
Molecular Docking: Autodock



The force field includes six pair-wise evaluations (V) and an estimate of the conformational entropy lost upon binding (ΔS_{conf}):

$$\Delta G = (V_{bound}^{L-L} - V_{unbound}^{L-L}) + (V_{bound}^{P-P} - V_{unbound}^{P-P}) + (V_{bound}^{P-L} - V_{unbound}^{P-L} + \Delta S_{conf})$$

where L refers to the "ligand" and P refers to the "protein" in a ligand-protein docking calculation.

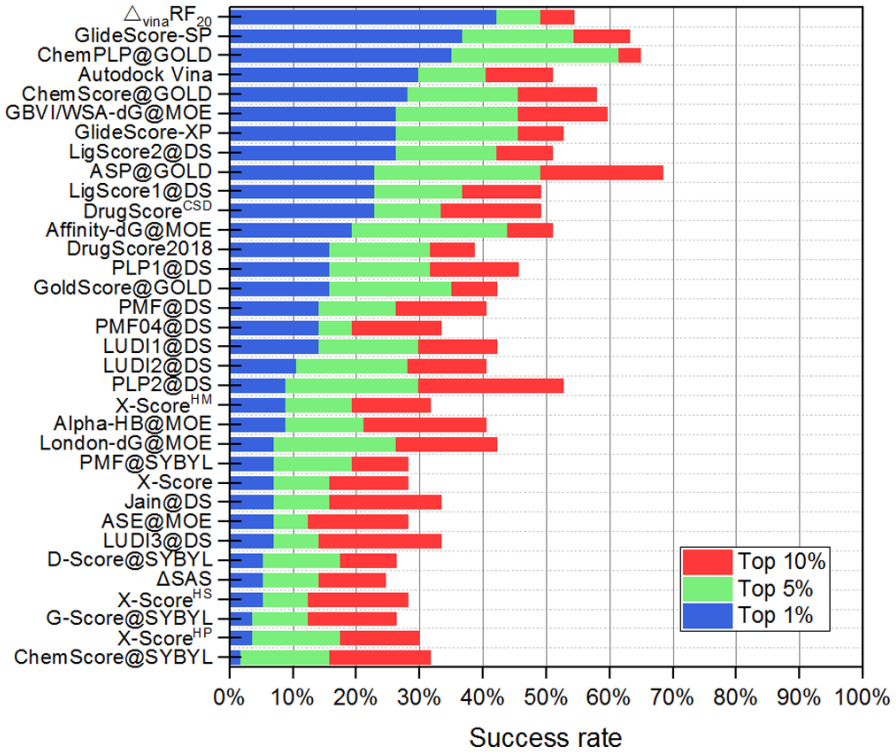


Each of the pair-wise energetic terms includes evaluations for dispersion/repulsion, hydrogen bonding, electrostatics, and desolvation:

$$V = W_{vdw}\sum_{i,j}\left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^{6}}\right) + W_{hbond}\sum_{i,j}E(t)\left(\frac{C_{ij}}{r_{ij}^{12}} - \frac{D_{ij}}{r_{ij}^{10}}\right) + W_{elec}\sum_{i,j}\frac{q_i q_j}{e(r_{ij})r_{ij}} + W_{sol}\sum_{i,j}\left(S_i V_j + S_j V_i\right)e^{(-r_{ij}^2/2\sigma^2)}$$

Autodock potential is based on Physics, but it introduce weighting pre-factor to each term (W) to better fit empirical data.

Example: Autodock Vina

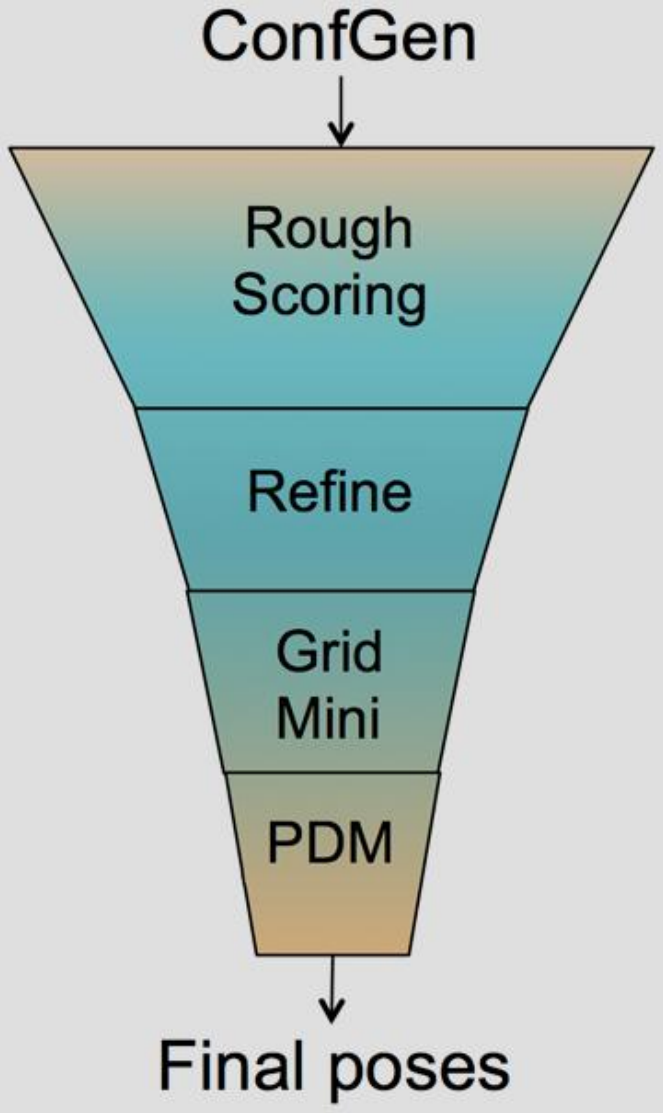


Success rate for detecting the highest-affinity ligand for a given target protein by each scoring function in the forward screening test. Blue, green, and red bars indicate the success rates obtained by considering the top 1%, 5%, and 10% candidates in screening, respectively. All scoring functions are ranked in a descending order by their success rates obtained at the top 1% level.



Su et al, JCIM, 2019

Example: Schoedinger and Glide



Schoedinger is a software that use an empirical scoring function to determine the best mode and evaluate the binding energy of the docking.

The general form of the function is public, but the coefficients and the implementation are proprietary and not made public.

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$$\Delta G_{\text{bind}} = C_{\text{lipo-lipo}} \sum f(r_{\text{lr}}) + C_{\text{hbond-neut-neut}} \sum g(\Delta r)\, h(\Delta\alpha) + C_{\text{hbond-neut-charged}} \sum g(\Delta r)\, h(\Delta\alpha) + C_{\text{hbond-charged-charged}} \sum g(\Delta r)\, h(\Delta\alpha) + C_{\text{max-metal-ion}} \sum f(r_{\text{lm}}) + C_{\text{rotb}} H_{\text{rotb}} + C_{\text{polar-phob}} V_{\text{polar-phob}} + C_{\text{coul}} E_{\text{coul}} + C_{\text{vdW}} E_{\text{vdW}} + \text{solvation terms}$$

Friesner et al., Journal of Medicinal Chemistry 47:1739-49 (2004)

How do we use scoring functions

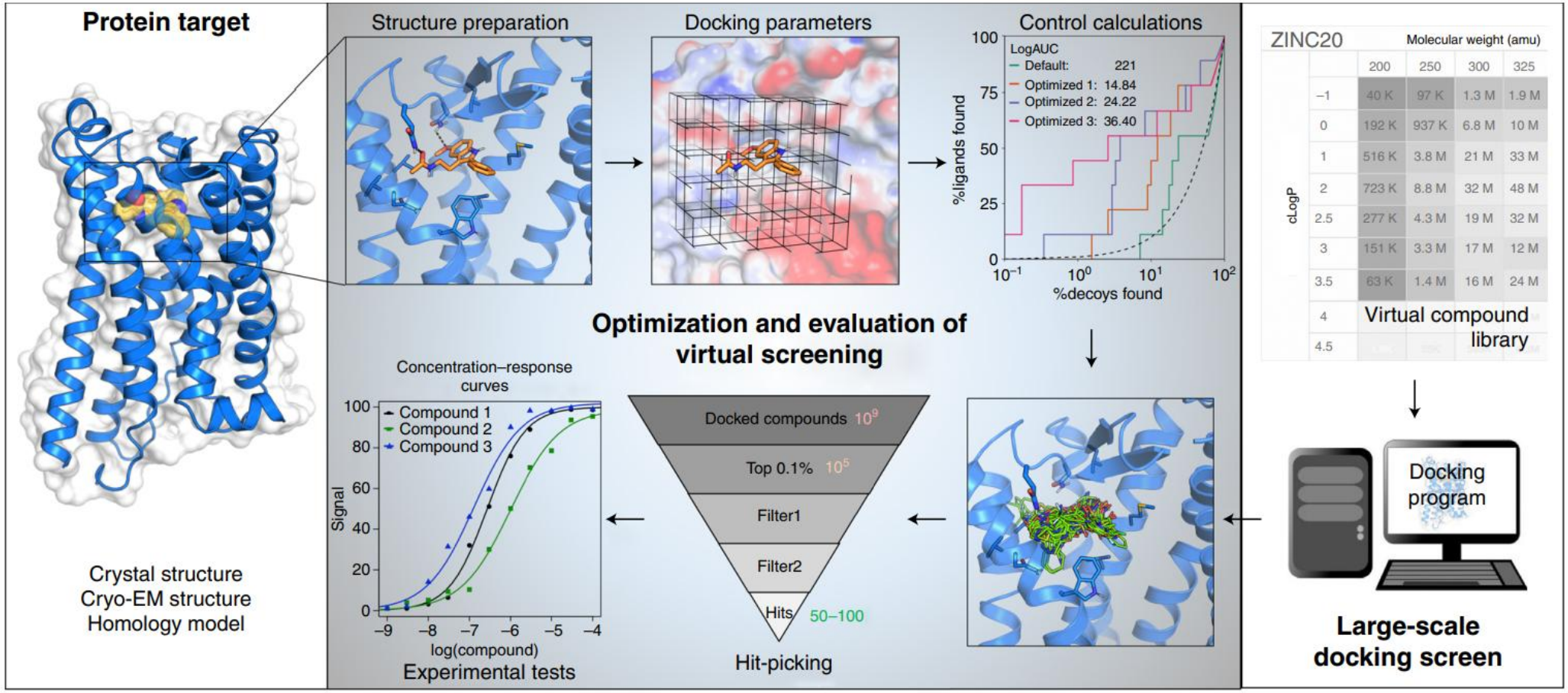
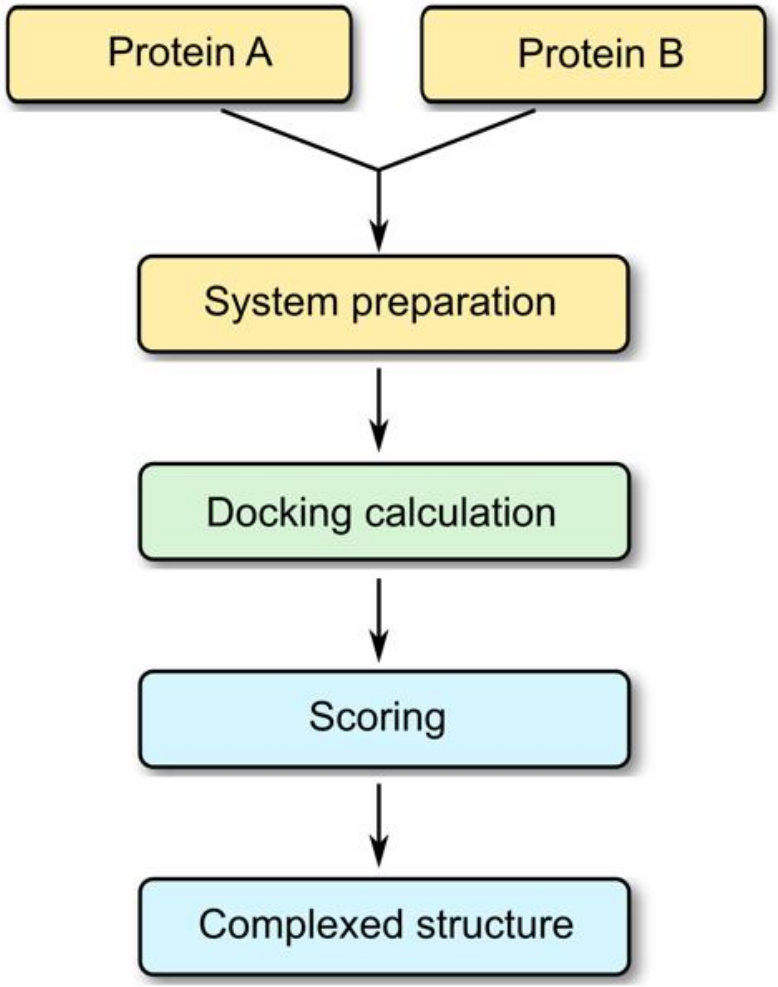
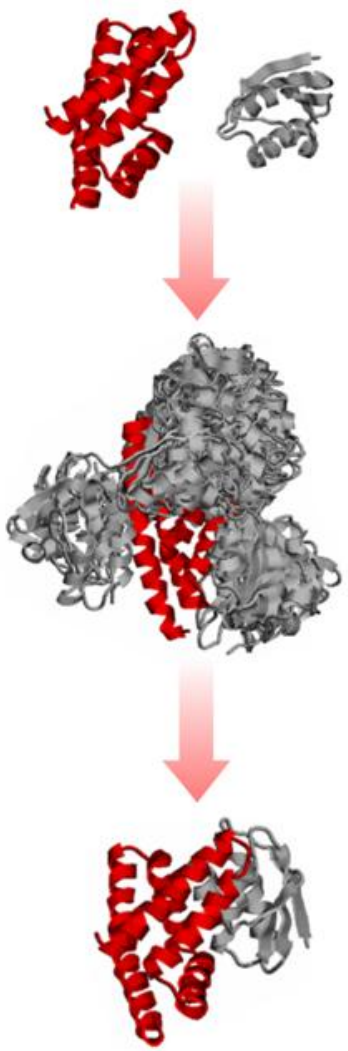


Fig. 1 | Large library docking workflow. The two required inputs for such a screen are the target structure and a screening database. Prior to using the database, the target structure must be converted into a representation used by the docking software and the pocket should be optimized with control calculations using retrospective analysis on known actives. After the prospective library has been docked, top-ranked hits can be filtered and selected for experiment. Multiple assays and controls are typically necessary to confirm activity.

Bender et. al. Nature Protocols 2021

Protein-Protein Docking





There are many Protein Docking methods, but they mostly follow the same strategy.

One protein is docked to the second using different orientation and translation (as a rigid body).

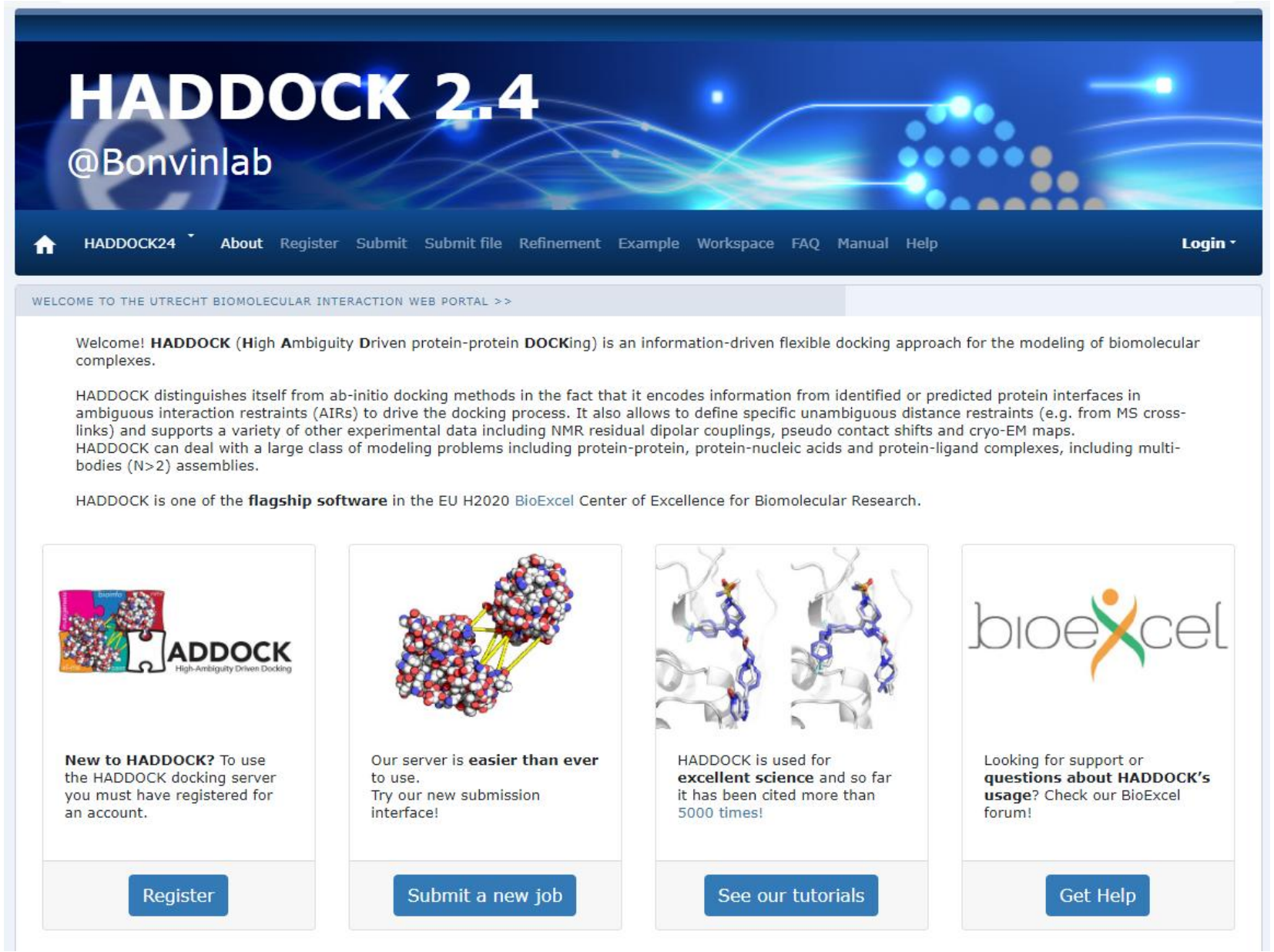
The docking is then refined allowing some flexibility of the side chains.

The various docking are scored according to a scoring function (which works like an energy function).

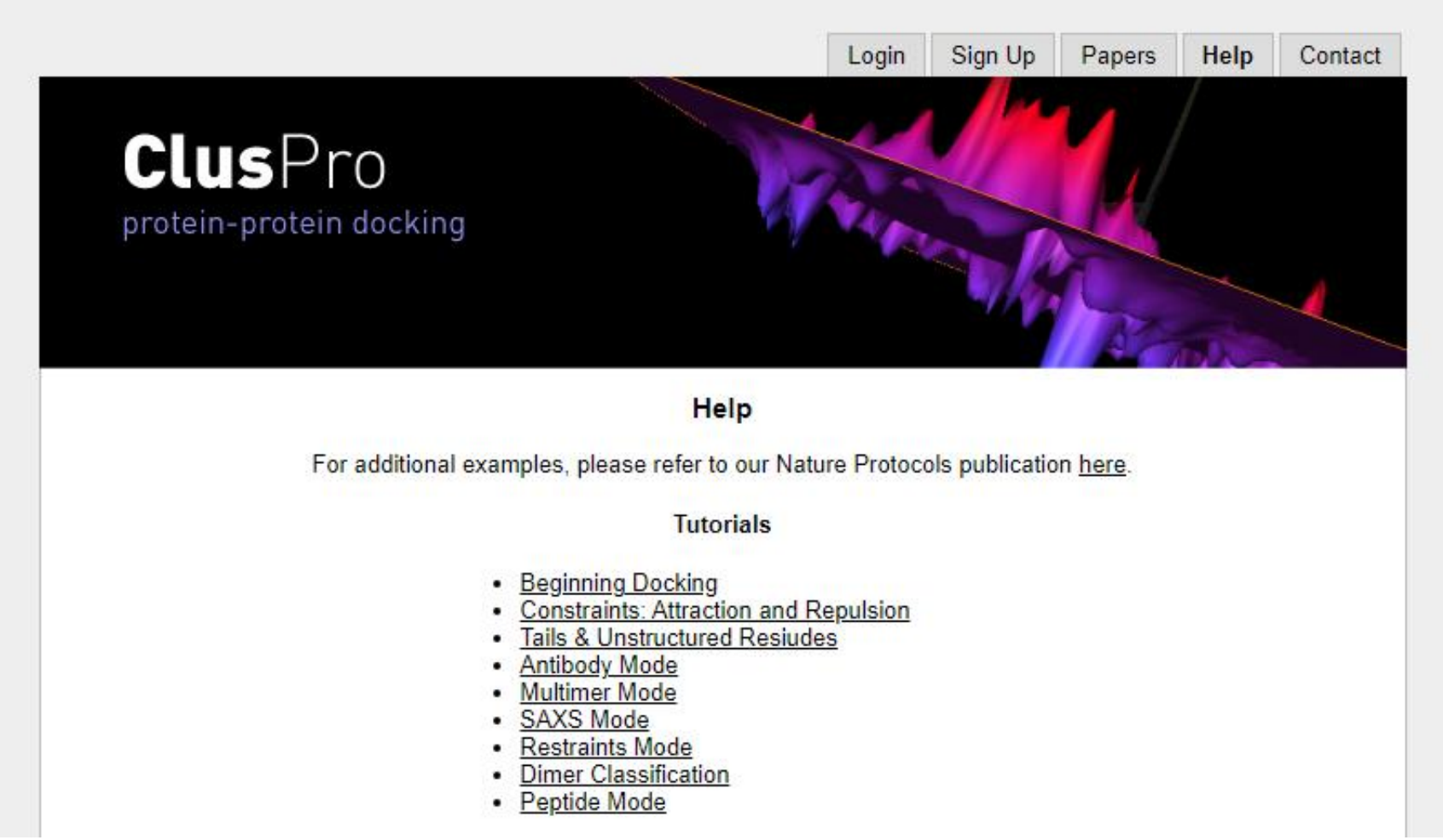
Similar docking can be grouped together to create a cluster, the cluster size can also be used as an approximation of the entropy.

The best docking are proposed and need to be evaluated.

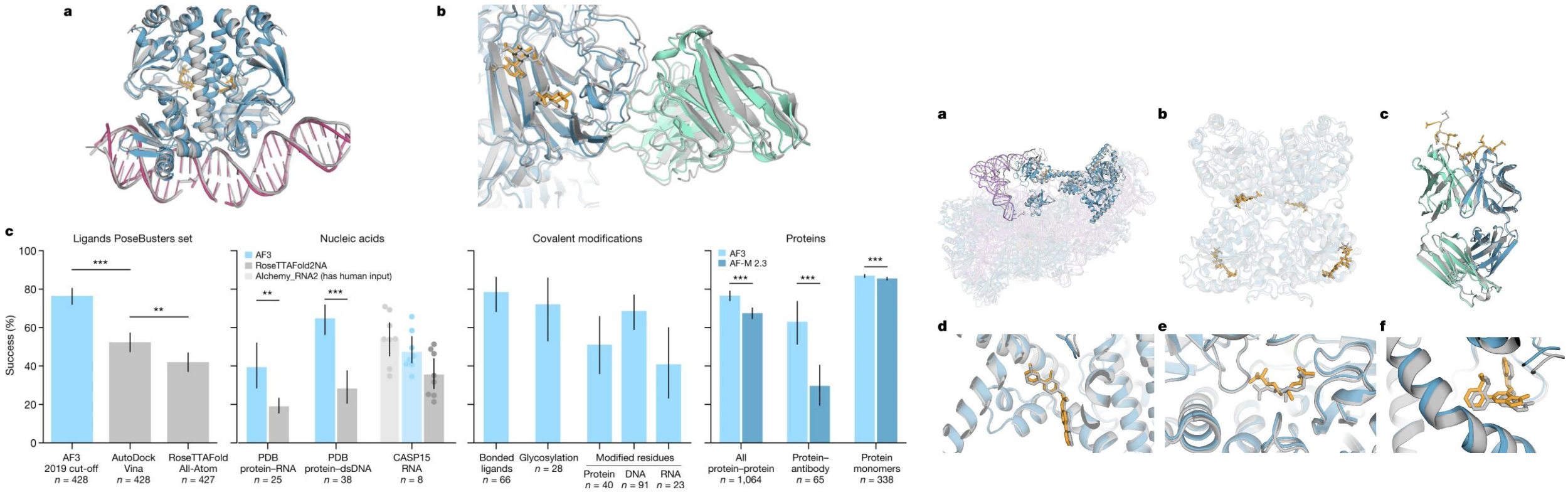
Protein – Protein: Haddock



Protein – Protein: ClusPro

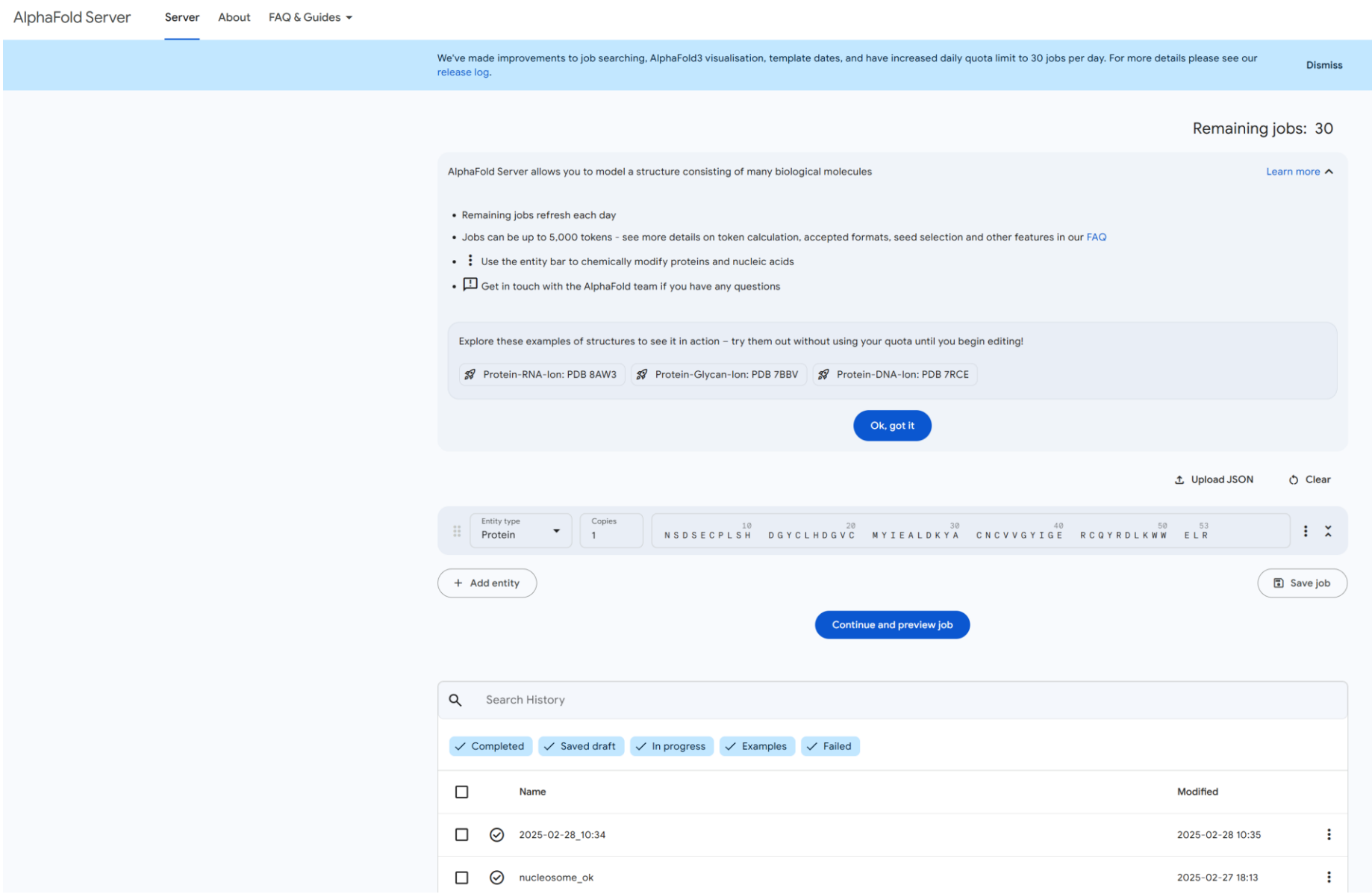


AF3 accurately predicts structures across biomolecular complexes



Selected structure predictions from AF3. Predicted protein chains are shown in blue (predicted antibody in green), predicted ligands and glycans in orange, predicted RNA in purple and the ground truth is shown in grey. **a**, Human 40S small ribosomal subunit (7,663 residues) including 18S ribosomal RNA and Met-tRNA$_i^{Met}$ (opaque purple) in a complex with translation initiation factors eIF1A and eIF5B (opaque blue; PDB 7TQL; full-complex LDDT, 87.7; GDT, 86.9). **b**, The glycosylated globular portion of an EXTL3 homodimer (PDB 7AU2; mean pocket-aligned r.m.s.d., 1.10 Å). **c**, Mesothelin C-terminal peptide bound to the monoclonal antibody 15B6 (PDB 7U8C; DockQ, 0.85). **d**, LGK974, a clinical-stage inhibitor, bound to PORCN in a complex with the WNT3A peptide (PDB 7URD; ligand r.m.s.d., 1.00 Å). **e**, (5S,6S)-O7-sulfo DADH bound to the AziU3/U2 complex with a novel fold (PDB 7WUX; ligand r.m.s.d., 1.92 Å). **f**, Analogue of NIH-12848 bound to an allosteric site of PI5P4Kγ (PDB 7QIE; ligand r.m.s.d., 0.37 Å).

AlphaFold 3


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Protein-Glycan-Ion: PDB 7BBV
Protein-DNA-Ion: PDB 7RCE
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Upload JSON
Clear
Entity type
Protein
Copies
1
NSDSECPLSH DGYCLHDGVC MYIEALDKYA CNCVVGYIGE RCQYRDLKWW ELR
Add entity
Save job
Continue and preview job
Search History
Completed
Saved draft
In progress
Examples
Failed
Name
Modified
2025-02-28_10:34
2025-02-28 10:35
nucleosome_ok
2025-02-27 18:13

CAPRI competition for protein-protein docking predictions

